# Graduation Project Report Template

**Department of Software Engineering**

**This document is prepared by the Software Engineering department. Its purpose is to help CS IT and Software Engineering students compile their undergraduate project report document in doing their final year project.**

**7/1/2022**

©JJU Department Software Engineering, Undergraduate Project Report Template Document

**Cover Page**

**JIGJIGA UNIVERSITY INSTITUTE OF TECHNOLOGY**
**DEPARTMENT OF SOFTWARE ENGINEERING**

**Undergraduate Project**

# Title: Title of your project

**Group Members:**

| No. | Name | ID |
|---|---|---|
| 1. | …….. | …….. |
| 2. | …….. | …….. |
| 3. | ……. | …….. |
| 4. | …….. | ……. |
| 5. | …….. | …….. |

**Advisor Name:_____Signature_____**

**dd/mm/yy**

©JJU Department Software Engineering, Undergraduate Project Report Template Document

## Acknowledgement

This is optional, although most reports include a brief statement of thanks in recognition of special assistance and guidance given by individuals, institutions or government bodies.

©JJU Department Software Engineering, Undergraduate Project Report Template Document

## Table of contents

The titles of sections, chapters and their principal subdivisions along with the page numbers on which they appear should be listed in the Table of Contents. Titles should be worded exactly as they appear in the text of the report. Reports with many subdivisions should use a hierarchical numbering system for headings and sub-headings (e.g., 3.1). Such a numbering system combined with the judicious use of upper and lower case, indentations and italics should provide a summary of the relationships between the sections of the report. The table of contents should be generated automatically. The following is a recommended content structure.

**Acknowledgments**

**Table of Contents**

**List of Tables and Figures Abstract**

**Chapter One: Introduction**

1.1 Background of the organization
1.2 Statement of the problem
1.2.1 Existing system
1.2.2 Major problem of existing system
1.2.3 Proposed system
1.2.4 Advantage of proposed system
1.3 Motivation
1.4 Scope and limitation of the project
1.4.1 Scope of the project
1.4.2 Limitation of the project
1.5 Objective
1.5.1 General objective
1.5.2 Specific objective
1.6 Methodology

1.6.1 Data collection methodology
1.6.2 System design and analysis tools
1.6.3 System development tools

**Chapter Two: System Requirement Specification**

2.1 Background Overview
2.2. Functional Requirement
2.3 Non-Function Requirement
2.4 Feasibility study

**Chapter Three: System Analysis and Modeling**

3.1 Overview
3.2. Scenario Based Modeling
3.2.1 Use case identification
3.2.2 Actor identification
3.2.3 Use case description
3.2.4 Activity diagram
3.3. Behavioral/Dynamic Modeling
3.3.1. Sequence diagram
3.3.2. State Diagram
3.4. Class-Based Modeling
3.4.1. Identifying classes
3.4.2. Class Diagram

**Chapter Four: System Design**

4.1 Overview
4.3 System Design
4.4. Architecture of the system
4.5. Database design
4.5 User Interface Design

**Chapter Five: Implementation and Testing**

5.1. Implementation

5.2. Testing

**Chapter Six: Conclusion and Future Work References**

**Appendix**

## Lists of Tables, Figures/Illustrations, Plates/Photographs

These lists consist of the exact titles (including numbering) of all tables, figures and plates that appear in the report. All tables, figures and plates should be numbered consecutively throughout the text.

©JJU Department Software Engineering, Undergraduate Project Report Template Document

## List of Abbreviations, Symbols, Specialized Nomenclature

This list is optional, depending on the subject of the report. All scientific symbols and nomenclature should follow the standard SI-system.

©JJU Department Software Engineering, Undergraduate Project Report Template Document

## Abstract

An abstract in brief English are required. The abstract is a summary of the entire report. It should briefly outline the research problems addressed by the report, the findings, and the significance of the work in the context of the field of study. The abstract should not exceed one typewritten single-spaced page of text (c. 300-400 words) with the font size of 11- points. Abstracts should be in English and should be italic style.

©JJU Department Software Engineering, Undergraduate Project Report Template Document

## *Main Text of the Report*

The main body of the report is usually arranged into consecutively numbered chapters or sections. The structure presented here should be adhered to as far as possible. However, the importance of each chapter (size and content) is likely to vary significantly from project to project, given the variety of projects undertaken. Therefore, in some cases it may be appropriate to devote several chapters to a particular topic or to add/remove chapters on topics not mentioned below.

## Chapter 1: Introduction

This chapter is concerned with similar concerns to the abstract and should provide an overview of the report with more detail. The methodology should be stated and describe here as different section. The contribution of the student should be stated precisely and in detail. The structure of the report should be given detailing where each chapter fits within the overall and what each contributes; it may be useful to provide a diagram showing dependencies and relationships between chapters.

Chapter 1 is often the last chapter to be written, and to be checked carefully before submission. It is a key chapter, which the external examiner may read when making his assessment.

# Chapter 2: System Requirement Specification

## Background (Overview)

This chapter should first present the background to the area of investigation and establishing the context of the problem. Understanding the problem domain in detail should be the first step and it is a foundation for the rest of the project work. Often this background consists of a survey and review of literature/documents associated with the problem context. Sometimes on site observation might be necessary.

The literature should not simply be represented but **critically analyzed** by the student. All sources used should be precisely cited in the text, i.e. in a way that enables the reader to access the source. If material is copied directly then it should be placed in quotes and the reference given quoted. In general, copying of this type should be avoided. If a diagram is copied then the caption should give the reference.

## Requirements

The major concern of this chapter is establishing the detailed requirements specification for the work. The nature of these requirements depends on the type of project being investigated. These requirements could be obtained from a number of sources (many of which may have been discussed in detail in the background). The chapter should indicate the ways in which the requirements have been obtained. Once obtained the requirements should be expressed, prioritized and detailed in an appropriate form. As much as possible the requirements should be measurable and quantified so that it can be determined if they have been met.

Requirements should be classified as functional and non-functional requirements. You should follow and apply Software Engineering practices for user requirements elicitation and engineering. Please refer your **Software Engineering course** on requirements elicitation and engineering.

# Chapter 3: Requirement Analysis and Modeling

Requirements analysis is an elaboration of the basic requirements established during requirement elicitation. This is the first technical representation of the proposed system using a series of requirements modeling approaches like,

- Scenario-based modeling (Use cases, Use case Diagrams, Activity Diagrams )
- Behavioral modeling (State Diagram, Sequence Diagram)
- Class-based modeling (Class Diagrams)

Once you identify your system requirements in chapter 2, you should elaborate it using the above requirements modeling approaches. This chapter is a basis for the design phase.

## Chapter 4: System Design & Architecture

The design is concerned with presenting the design of the artifact developed and justifying how it should meet the identified requirements. It is likely to consist of three parts:

1. How and why the design has been carried out – the approach and notation used, etc;
2. The conceptual level design;
3. Communication and description of the design.

As appropriate, alternatives considered may be discussed with justification for the approach taken. The design should be expressed and detailed in a suitable form. Also, this chapter should include database design and user interface design in detail.

## Chapter 5: Implementation and Testing

This chapter focuses on the realization of the design by an implementation. The behavior of the implementation should be described and a justification of why it satisfies the design should be given. The actual implementation **should** be described but not usually in great detail; it is rare that all code will be included. In general, only code that highlights particular approaches or represents important components should be included.

It is often appropriate (and easiest) to provide a walk-through of the system to explain its behavior and relate it to the design. In certain cases a user manual may be presented in the appendix and this can be referred to here. At each stage, evidence and justification should be produced. This can be obtained from a number of sources; if the aim was to improve performance or accuracy then it is relatively easy to evaluate; in other cases the evaluation may be via a user set, this is often of particular importance in projects what involve user interfaces, web-based activity, etc.

The environment of all tests should be fully specified, i.e. so that the reader could recreate the tests and results given a similar environment. As minimum requirement, Usability and Functional Testing include test plan should be produced here. Depending on the type of project, it may be necessary to show how the solution produced – usually small and prototypical – would scale up to a real-world system.

## Chapter 6: Conclusion & Further Work

This chapter should present conclusions about the investigation and outline further work. This chapter should not be left until the end of the project period. Valuable ideas should be collected throughout the project and added to a chapter outline. The chapter should re-outline what has been done in the investigation, and been shown in the report. The lessons learned from the overall investigation should be presented with appropriate examples.

The evaluation together with new ideas should naturally lead to further work that would “improve” the work in some sense. The further work section should be substantial in that this is an important part of a scientific investigation. Often the depth of further work is a good indication of how well aware the student is of the topic of investigation.

©JJU Department Software Engineering, Undergraduate Project Report Template Document

## References

Any report, which makes use of other works, either in direct quotation or by reference, must contain references listing all of these sources. Only works directly cited or quoted in the text should be included in the references. How to use and list the references is already covered on ***proposal template*** document.

## Appendices

This section is optional and will depend on the individual report content. It contains supplementary illustrative material, original data, and quotations too long for inclusion and not immediately essential to an understanding of the subject. This section may be divided into sections as Appendices A, B, C, etc. Any figures or tables included in the appendix should be numbered and captioned as for all text tables and figures.

©JJU Department Software Engineering, Undergraduate Project Report Template Document

***Note:***

The internal organization of the report is the responsibility of the Candidate in consultation with his/her supervisor(s). The organization will partly depend on the field of study, but the responsibility is on the student to provide a systematic and well-organized report. Overall, the font of the main text should be specified on the guideline. In addition, candidate should put a header for each page, which is the title of the chapter concerned.

©JJU Department Software Engineering, Undergraduate Project Report Template Document